

AeDAM: An Event-Driven Architecture Mapping exploration tool

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AeDAM is a design-space and mapping exploration framework purpose-built for event-driven neuromorphic accelerators. It replaces frame-based assumptions with event-driven mapspace generation and an analytical cost model for asynchronous word-access behavior, enabling accurate energy, latency, and area estimates for various AI workloads.

Validated on SENECA, AeDAM delivers ~2.5× faster mapping exploration and 13–52% lower latency mappings when benchmarked against SoTA DSE tools like Zigzag.

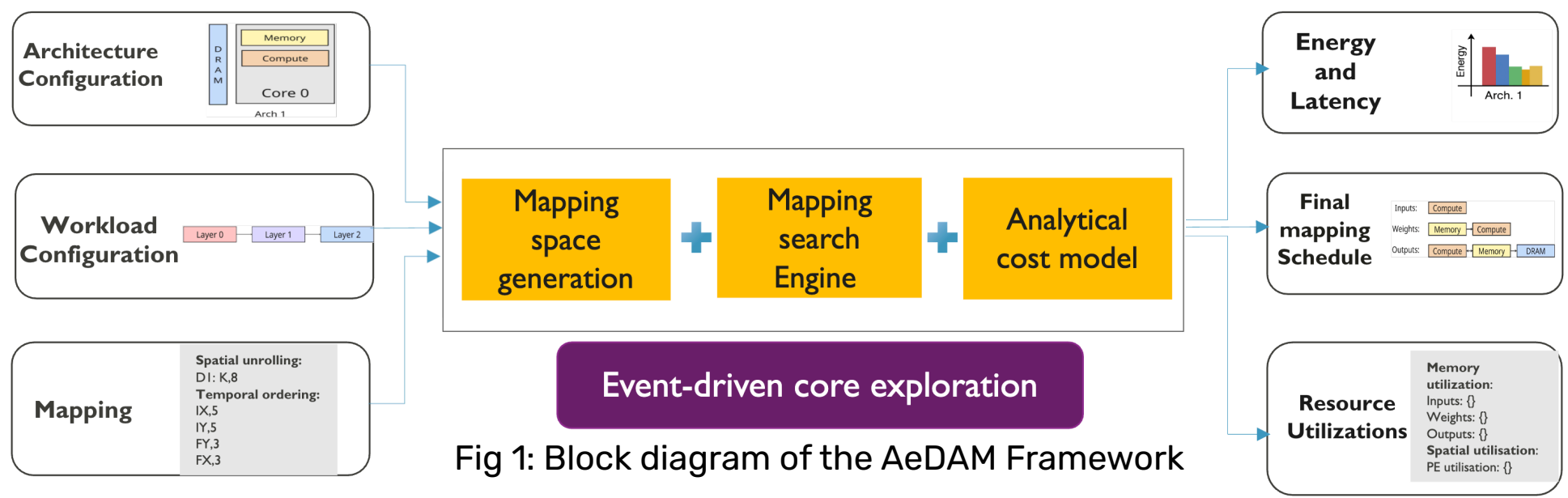
01 PROBLEM STATEMENT

- Mapping tools built for synchronous, frame-based accelerators are inefficient for event-driven architectures.
- Existing DSE flows can't capture asynchronous execution, irregular timing, and sparsity-driven access that define event-driven systems.

02 RELATED WORKS

- Timeloop, ZigZag, ConvFusion, Sparseloop, Stream: powerful, but all assume traditional, synchronous execution.
- None model event-driven architectures behaviour; this is the gap AeDAM targets.

03 AeDAM FRAMEWORK

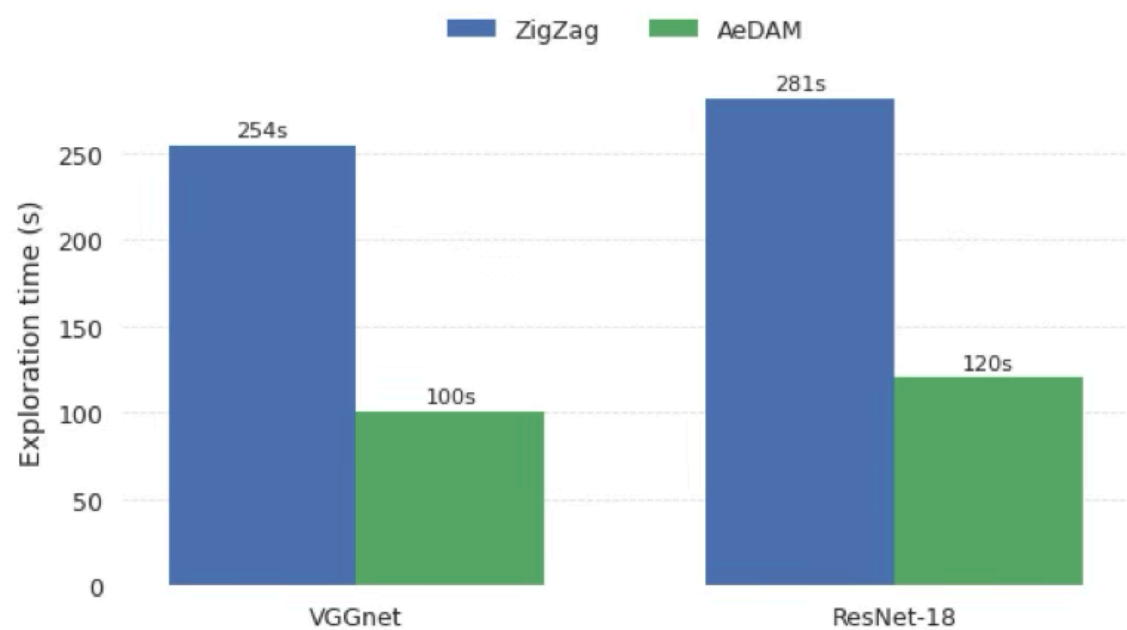


- Specialized workflow for event-driven accelerators: Given an architecture + workload, AeDAM builds an event-driven mapspace, evaluates candidates with asynchronous-aware cost models, and outputs the best mapping schedule with energy/latency/area metrics and resource-utilization.
- Dual use (mapping + architecture DSE): Finds the best mapping for a specific AI workload and performs design-space exploration of the event-driven architecture for that workload (e.g., SRAM capacity/bandwidth, PE-array organization) to identify Pareto-optimal configurations.

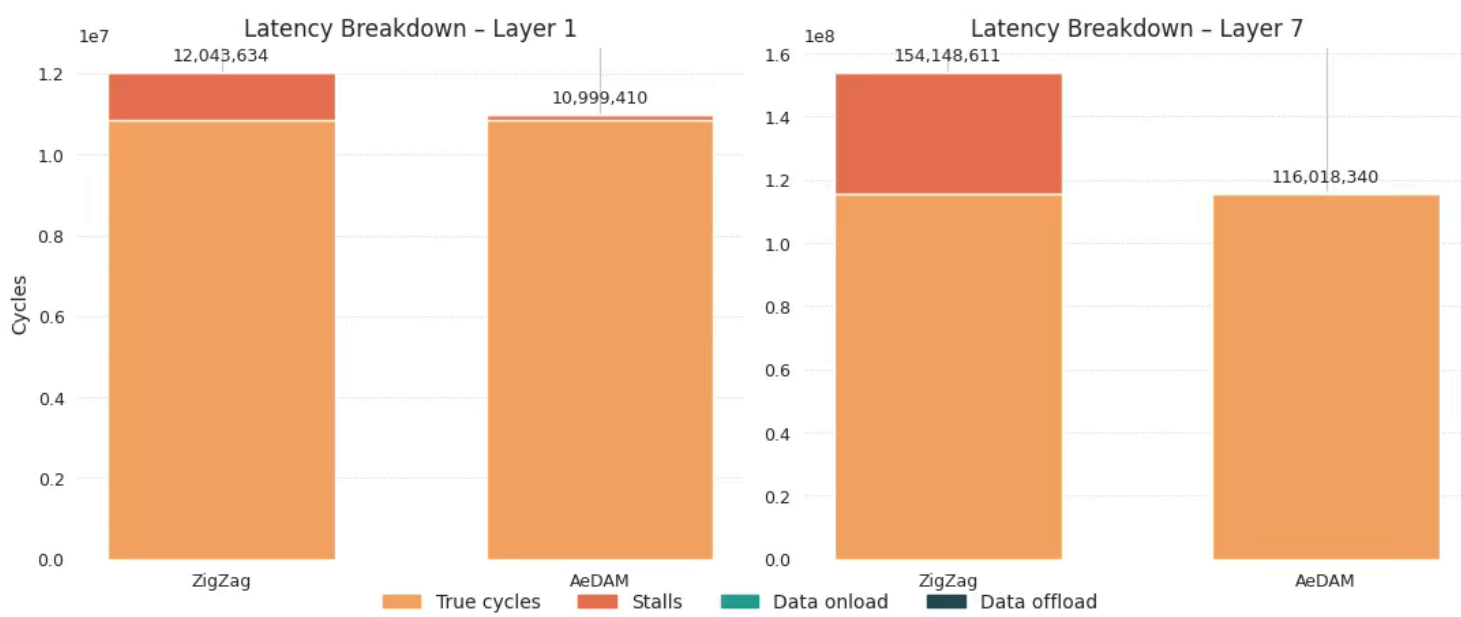
04 FRAMEWORK HIGHLIGHTS

- Event-driven mapspace generation via LOMA transformation + heuristic pruning (LPF grouping).
- The analytical cost model is extended for asynchronous word access, with detailed latency, energy, and area models estimates for AI workload processing.
- Unified exploration producing best mapping schedules and utilization/metric reports.

05 RESULTS



- Exploration speed: ~2.5× faster than ZigZag for VGGNet (100s vs 254s).



- Latency: 13% better (Layer-1) and 52% better (Layer-7) vs ZigZag.

- Design study (SENECA): 2.96× latency reduction, 1.37× energy improvement, 12× E×D² reduction at the optimal config.

06 CONCLUSION & FUTURE WORKS

- AeDAM addresses the core inefficiency of applying synchronous mapping tools to event-driven accelerators and outperforms state-of-the-art frameworks.
- Provides a specialized framework for systematic exploration of event-driven architectures.
- Extend to multi-core event-driven dataflow; add weight/input sparsity & adaptive sparsity; support SNNs.